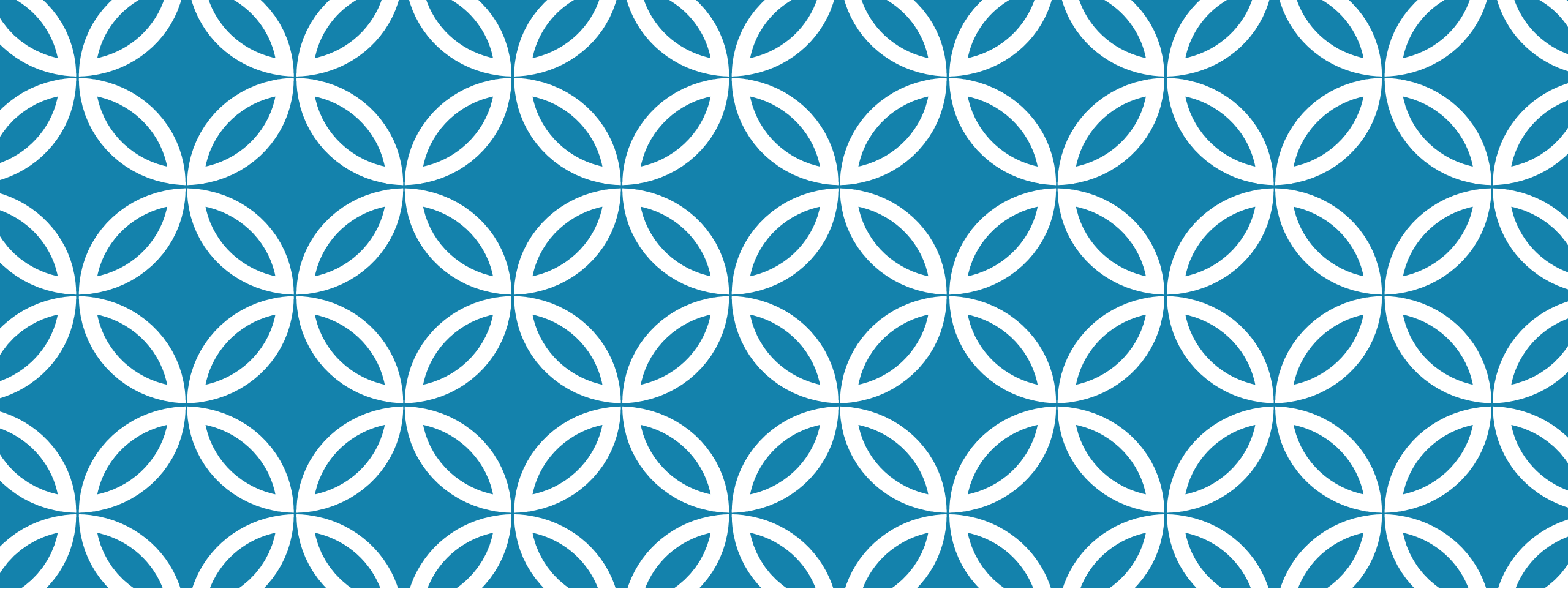




CSE455 - SOFTWARE TESTING

Chapter 1

COURSE OUTLINE

Introduction

Software testing basics

Testing throughout the software development life cycle

Static Testing

Dynamic Testing

Test Management

Testing tools

HIGH DEPENDENCY ON SOFTWARE SYSTEM

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Software and software quality therefore not only govern how large parts of our lives function, they are also increasingly important factors in our everyday safety.

Smooth functioning of countless companies relies on reliability of software systems.

IMPORTANCE OF TESTING

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Improve overall software quality

CERTIFICATIONS IN SOFTWARE TESTING

ISTQB (International Software Testing Qualifications Board) coordinates qualification activities in individual countries and ensures the global consistency and comparability of the syllabi and exam papers

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Basics of Software Testing are covered in *Foundation Level*, you can further move to *Advance Level* exam, which offers deep understanding of evaluation and testing techniques. The *Expert Level* certificate is aimed at experienced software testing professionals.

CASE STUDY: VIRTUALSHOWROOM VSR-II

Electronic sales system managed by a car manufacturer

Five modules

DreamCar: Customers can configure their vehicle (model, color, extras, and so on) and calculate price

EasyFinance: for optimal financing

JustInTime: to order the vehicle

NoRisk: Select insurance

FactBook: manages all customer and contract data

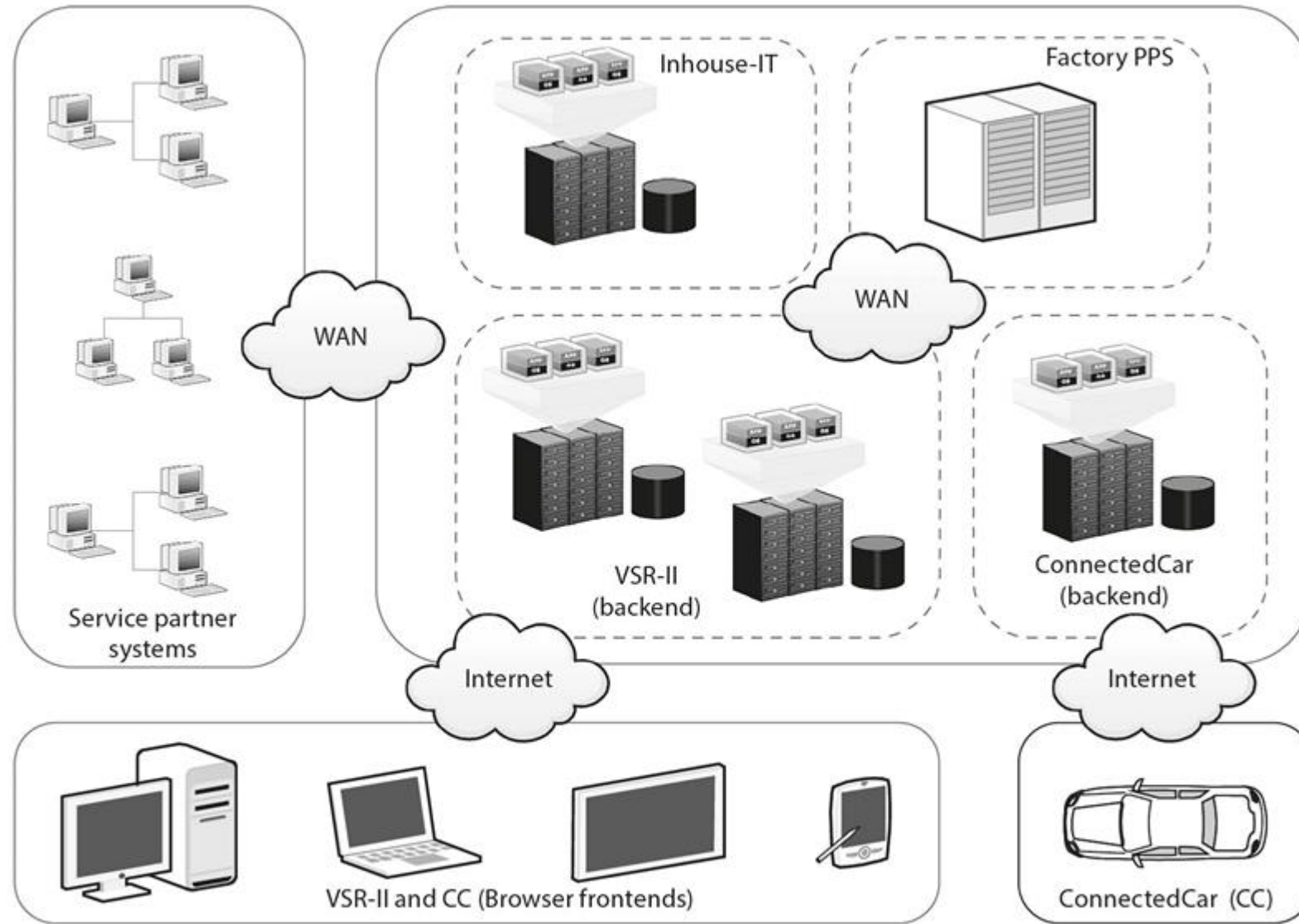


Fig. 1-1 VSR-II overview

CONT.

The manufacturer's sales and marketing department has decided to update the system and has defined the following objectives

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VSR is a traditional client-server system. The new VSR-II will be web based and will be accessible on all types of devices

The *DreamCar*, *EasyFinance*, *FactBook*, *JustInTime*, and *NoRisk* modules will be ported to the new technology

CONT.

The new *ConnectedCar* module is to be integrated into the system. This module collects and manages status data for all vehicles sold, and communicates data relating to scheduled maintenance and repairs to the driver as well as to the dealership and/or service partner.

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Each of the five existing modules will be ported and developed by a dedicated team. An additional team will develop the new *ConnectedCar* module. The project employs a total of 60 developers and other specialists from internal company departments as well as a number of external software companies.

CONT.

The teams will work using the *Scrum* principles of agile development. This agile approach requires each module to be tested during each iteration. The system is to be delivered incrementally.

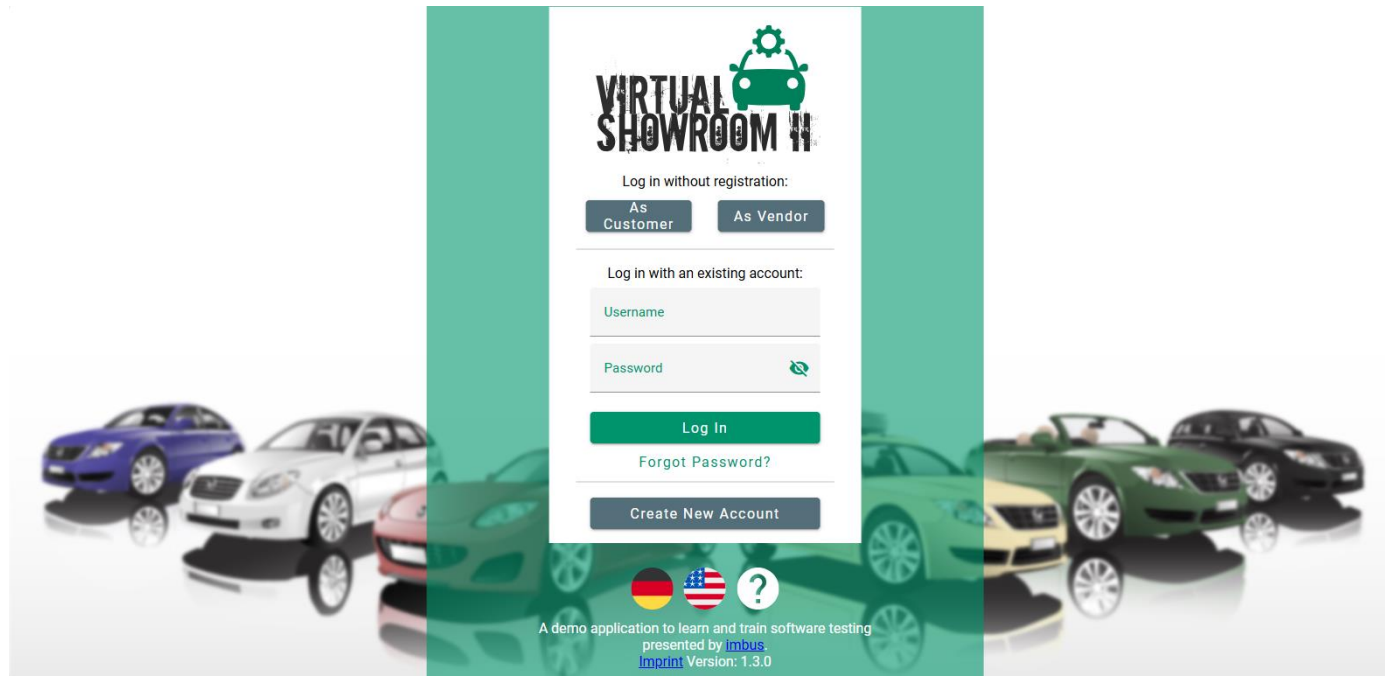
CONT.

The teams will work using the *Scrum* principles of agile development. This agile approach requires each module to be tested during each iteration. The system is to be delivered incrementally.

In order to avoid complex repeat data comparisons between the old and new systems, *VSR-II* will only go live once it is able to duplicate the functionality provided by the original *VSR* system

CONT.

1. Visit <http://vsr.testbench.com/>
2. Create your personal VSR-II training workspace
3. Log into your VSR-II workspace and start



CONCLUSION

- Testing ensures software reliability and quality
- Different certifications at 3 stages are available
- VSR is a suitable platform for software testing